Example	$L_i^- - L_i^+$	Useful
The Flodden Window (a war memorial dedicated to The Middleton Archers), in the Grade I-listed Church of St Leonard in Middleton is said to be the oldest war memorial in the United Kingdom. <API> **WikiSearch(War memorial Flodden) → Battle of Flodden > Commemoration > The stained-glass Flodden Window in Middleton Parish Church [...] was constructed by Sir Richard Assheton in memory of the Battle of Flodden and the archers from Middleton who fought in it.** </API> Sir Richard Assheton of Middleton (who built St Leonard) was granted knighthood [...]	5.49	✓
Note: The WL will be open on Friday, <API> **Calendar() → Today is Thursday, March 9, 2017.** </API> March 10, and Sunday, March 19 for regular hours.	2.11	✓
The Nile has an approximate length of <API> **QA(What is the approximate length of the Nile?) → 6,853 km** </API> 6,853 kilometers, the White Nile being its main source.	2.08	✓
If Venus had an atmosphere similar to Earth's then you would expect Venus' mean temperature to be 499 K (1.74 x 287) rather than 735 K which is <API> **Calculator(735 / 499) → 1.47** </API> 1.47 (735 / 499) times hotter than it should be.	1.59	✓
You are here: Home / Featured / Catch this fast train to success! <API> **WikiSearch(Fast train success) → Fast Train > It also peaked at #23 on the Canadian CHUM singles chart, on June 26, 1971. The success of this single established Myles Goodwyn as the band's main songwriter, and made it possible for April Wine to record a second album.** </API> Don't wait weeks and [...]	0.92	✗
Os Melhores Escolas em Jersey 2020 <API> **MT(Os Melhores Escolas em Jersey) → The Best Schools in Jersey** </API> On this page you can search for Universities, Colleges and Business schools in Jersey	0.70	✓
Enjoy these pictures from the <API> **Calendar() → Today is Friday, April 19, 2013.** </API> Easter Egg Hunt.	0.33	✓
85 patients (23%) were hospitalised alive and admitted to a hospital ward. Of them, <API> **Calculator(85 / 23) → 3.70** </API> 65% had a cardiac aetiology [...]	−0.02	✗
But hey, after the <API> **Calendar() → Today is Saturday, June 25, 2011.** </API> Disneyland fiasco with the fire drill, I think it's safe to say Chewey won't let anyone die in a fire.	−0.41	✗
The last time I was with <API> **QA(Who was last time I was with?) → The Last Time** </API> him I asked what he likes about me and he said he would tell me one day.	−1.23	✗

Table 10: Examples of API calls for different tools, sorted by the value of $L_i^- - L_i^+$ that is used as a filtering criterion. High values typically correspond to API calls that are intuitively useful for predicting future tokens.

approaches, additional information is *always* provided, regardless of whether it is helpful or not. In contrast, Toolformer learns for itself to explicitly asks for the right information.

Tool Use Several approaches aim to equip LMs with the ability to use external tools such as search engines (Komeili et al., 2022; Thoppilan et al., 2022; Lazaridou et al., 2022; Shuster et al., 2022; Yao et al., 2022), web browsers (Nakano et al., 2021), calculators (Cobbe et al., 2021; Thoppilan et al., 2022), translation systems (Thoppilan et al., 2022) and Python interpreters (Gao et al., 2022). The way these models learn to use tools can roughly be divided into two approaches: Either they rely on large amounts of human supervision (Komeili et al., 2022; Nakano et al., 2021; Thoppilan et al., 2022) or they work by prompting the language model in a few-shot setup tailored towards a specific task where it is known a priori which tools needs to be used (Gao et al., 2022; Lazaridou et al., 2022; Yao et al., 2022). In contrast, the self-supervised nature of Toolformer enables it to learn how and when to use tools without requiring a specific prompt that shows task-specific examples of how a tool could be used. Perhaps most closely related to our work is TALM (Parisi et al., 2022), an approach that uses a similar self-supervised objective for teaching a model to use a calculator and a search engine, but explores this only in settings where a model is finetuned for downstream tasks.

Bootstrapping The idea of using self-training and bootstrapping techniques to improve models has been investigated in various contexts, ranging from word sense disambiguation (Yarowsky, 1995), relation extraction (Brin, 1999; Agichtein and Gravano, 2000), parsing (McClosky et al., 2006; Reichart and Rappoport, 2007), sequence generation (He et al., 2020), few-shot text classi-

fication (Schick and Schütze, 2021a) and retrieval (Izacard and Grave, 2021) to reasoning (Zelikman et al., 2022). In a similar spirit to these approaches, Toolformer is trained on its own predictions after applying a perplexity-based filtering step.

7 Limitations

While our approach enables LMs to learn how to use a variety of tools in a self-supervised way, there are some clear limitations to what can be achieved with our method in its current form. One such limitation is the inability of Toolformer to use tools in a *chain* (i.e., using the output of one tool as an input for another tool). This is due to the fact that API calls for each tool are generated independently; as a consequence, there are no examples of chained tool use in the finetuning dataset. Our current approach also does not allow the LM to use a tool in an *interactive* way – especially for tools such as search engines, that could potentially return hundreds of different results, enabling a LM to browse through these results or to refine its search query in a similar spirit to Nakano et al. (2021) can be crucial for certain applications. Beyond this, we found models trained with Toolformer to often be sensitive to the exact wording of their input when deciding whether or not to call an API; this is perhaps unsurprising given that LMs are known to be very sensitive to the prompt they are provided with in both zero- and few-shot settings (Jiang et al., 2020; Schick and Schütze, 2021a). Depending on the tool, our method is also very sample-inefficient; for example, processing more than a million documents results in only a few thousand examples of useful calls to the calculator API. A potential solution to this problem might be to iteratively apply our approach, similar to how this is done in related bootstrapping approaches (Schick and Schütze, 2021a; Izacard and Grave, 2021; Parisi et al., 2022). Finally, when deciding whether or not to make an API call, Toolformer currently does not take into account the tool-dependent, computational cost incurred from making an API call.

8 Conclusion

We have introduced Toolformer, a language model that learns in a self-supervised way how to use different tools such as search engines, calculators, and translation systems via simple API calls. This is done by finetuning on a large number of sampled API calls that are filtered based on whether they reduce perplexity on future tokens. Toolformer considerably improves zero-shot performance of a 6.7B parameter GPT-J model, enabling it to even outperform a much larger GPT-3 model on a range of different downstream tasks.

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